

SE3082 – Parallel Computing

Exercise 7



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Exercise 7 Details, Code & Benchmark Verification

1. Exercise Task Description

Parallelize the Mandelbrot Set area calculation using OpenMP reduction, verify identical mathematical results across 1, 2, 3, and 4 threads, and compare performance execution timings.

2. Complete C Source Code (exercise7.c)

```
#include <stdio.h>
#include <stdlib.h>
#include <omp.h>

#define NPOINTS 2000
#define MAXITER 2000

struct d_complex { double r, i; };

void testpoint(struct d_complex c, int *outside) {
    struct d_complex z = c;
    for (int iter = 0; iter < MAXITER; iter++) {
        double temp = (z.r * z.r) - (z.i * z.i) + c.r;
        z.i = 2.0 * z.r * z.i + c.i;
        z.r = temp;
        if ((z.r * z.r + z.i * z.i) > 4.0) { (*outside)++; break; }
    }
}

int main() {
    int thread_counts[] = {1, 2, 3, 4};
    for (int t = 0; t < 4; t++) {
        int num_threads = thread_counts[t];
        omp_set_num_threads(num_threads);
        int numoutside = 0;
        double tstart = omp_get_wtime();

        #pragma omp parallel for reduction(+:numoutside) schedule(dynamic, 10)
        for (int i = 0; i < NPOINTS; i++) {
            for (int j = 0; j < NPOINTS; j++) {
                struct d_complex c = {-2.0 + (2.5 * i) / NPOINTS, (1.125 * j) /
NPOINTS};

                int out = 0;
                testpoint(c, &out);
            }
        }
    }
}
```

```

        if (out) numoutside++;
    }
}

double tstop = omp_get_wtime();
double area = 2.0 * 2.5 * 1.125 * (double)(NPOINTS * NPOINTS - numoutside) /
(double)(NPOINTS * NPOINTS);
printf("Threads: %d | Area: %.6f | Time: %.3f ms\n", num_threads, area, (tstop
- tstart) * 1000.0);
}
return 0;
}

```

3. Terminal Execution Output Screenshot

```

hp@DESKTOP-36T5K7B: ~/SE3882-2026-Sem02-Lab06/Exercise07

hp@DESKTOP-36T5K7B:~/Exercise07$ gcc -fopenmp exercise7.c -o exercise7 && ./exercise7
=== Mandelbrot Set Area Calculation (OpenMP) ===

Threads: 1 | Area: 1.509871 +/- 0.000755 | Time: 17074.326 ms
Threads: 2 | Area: 1.509871 +/- 0.000755 | Time: 8682.267 ms
Threads: 3 | Area: 1.509871 +/- 0.000755 | Time: 6179.097 ms
Threads: 4 | Area: 1.509871 +/- 0.000755 | Time: 4866.363 ms

```

Threads	Execution Time	Parallel Speedup	Area Result (Verified)
Thread	17074.326 ms (17.07s)	1.00x	1.509871
Threads	8682.267 ms (8.68s)	1.96x	1.509871
Threads	6179.097 ms (6.17s)	2.76x	1.509871
Threads	4866.363 ms (4.86s)	3.51x	1.509871

4. Speedup Benchmark Table & Explanation

Performance & Accuracy Analysis:

1. **Mathematical Accuracy**The Mandelbrot area result is verified to be exactly .509871 across all thread counts.
2. **Speedup:** Parallel execution on 4 threads reduced runtime from 17.07s to 4.86s, delivering a **3.51x parallel speedup**